A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Oscillations

Pendulums, guitar strings, car suspensions, the atoms in a crystal and the swaying of a skyscraper all oscillate. Most of them follow the same simple rule, simple harmonic motion, and once you understand that one rule you can predict their periods, their speeds and their energies. Then we look at what happens when oscillations are damped, and when something pushes them at just the right frequency.

SPEC 143 The condition for simple harmonic motion

WHAT THIS SPEC POINT ASKS

Understand that SHM needs a restoring force F = −kx, and use this to decide when SHM will happen.

Definition

An object moves with **simple harmonic motion** (SHM) when its **acceleration (or resultant force) is proportional to its displacement from the equilibrium position, and always acts in the opposite direction to the displacement** (towards the equilibrium position).

The condition for simple harmonic motion FORMULA SHEET

$$F = -kx \qquad \Rightarrow \qquad a = -\frac{k}{m}x$$

x: displacement from the equilibrium position / m k: constant (the stiffness, for a spring) / N m^{-1}

- x = displacement from the equilibrium position (m); k = a constant (for a spring, the stiffness, N m^{-1}).
- The **minus sign** means the force points the opposite way to the displacement: it is a **restoring** force.
- Since a = −(k/m)x and k/m is constant, comparing with a = −ω²x (144) gives **ω² = k/m**.

Spotting SHM

System	Restoring force	SHM?
Mass on a spring (within the limit of proportionality)	Hooke's law: F = −kx	yes

System	Restoring force	SHM?
Simple pendulum, **small** angles (under about 10°)	$F \approx -(mg/l)x$	approximately
A ball bouncing on the floor	weight, constant while in the air (not $\propto x$)	no
Pendulum swung through a large angle	not proportional to x	no
Liquid oscillating in a U-tube; a floating object bobbing	$\propto$ displacement	yes

- The test is always: write the resultant force (or acceleration) in terms of x. If it is "−(constant) × x", the motion is SHM.

WORKED EXAMPLE

Restoring force and acceleration

A 0.40 kg mass on a spring (k = 25 N m^{-1}) is pulled 3.0 cm below its equilibrium position and released. Find the force and acceleration at release.

$F = -kx = -25 \times 0.030 = -0.75$ N (magnitude 0.75 N, upwards, towards equilibrium)

$a = F/m = -0.75/0.40 =$ **-1.9 m s^{-2}** (1.9 m s^{-2} upwards)

MARK-SCHEME WORDING

Two marks, asked in 15 of 18 papers:

- "(resultant) force / acceleration is **proportional to the displacement** (from the equilibrium position)";
- "and acts in the **opposite direction to the displacement**" (or "(always) towards the equilibrium position").
- "Equilibrium **position**" (or "undisplaced / fixed / central point or position") is needed: "reject 'equilibrium'" on its own (WPH15 Jun 2026); "do not accept mean position" (WPH15 Oct 2022). One paper rejected "towards the equilibrium position" and wanted "opposite to the displacement" (WPH15 Oct 2025): use that wording to be safe.
- "An equation with symbols defined correctly is a valid response for both marks", but every term must be defined and the minus sign explained (WPH15 Jun 2023).

Often asked as: state the condition for SHM; explain how an a–x graph shows SHM; decide which situation is SHM.

COMMON MISTAKE

Writing "equilibrium" without "position". Saying "acceleration is proportional to displacement" and stopping (no direction). Writing a = −ω²x without saying what a, ω and x are or what the minus sign means. Repeating the learnt definition when the question says "explain how the **graph** shows SHM": refer to the graph (a straight line through the origin with a negative gradient).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- For an a–x graph question: "straight line **through the origin**" gives proportionality and "**negative gradient**" gives the opposite direction; the gradient is −ω² (WPH15 Jan 2025).

SPEC 144 The SHM equations

WHAT THIS SPEC POINT ASKS

Use a = −ω²x, x = A cos ωt, v = −Aω sin ωt, a = −Aω² cos ωt, T = 1/f = 2π/ω and ω = 2πf for a simple harmonic oscillator.

The equations

Defining equation of SHM

FORMULA SHEET

$$a = -\omega^2 x$$

ω: angular frequency / rad s^{-1}

Displacement

FORMULA SHEET

$$x = A\cos\omega t$$

A: amplitude / m

Velocity

FORMULA SHEET

$$v = -A\omega\sin\omega t$$

Acceleration

FORMULA SHEET

$$a = -A\omega^2 \cos \omega t$$

Period, frequency and ω

FORMULA SHEET

$$\omega = 2\pi f = \frac{2\pi}{T} \qquad T = \frac{1}{f}$$

f: frequency / Hz T: period / s

- A = **amplitude** (maximum displacement, m); ω = **angular frequency** (rad s^{-1}); f = frequency (Hz); T = period (s); t = time (s).
- x = A cos ωt assumes the timing starts at maximum displacement (released from x = A). If it starts at the equilibrium position moving in the +x direction, x = A sin ωt instead.
- v is the gradient of x–t, and a is the gradient of v–t (146, 147), which is why each line follows from the one before.
- Substituting x = A cos ωt into a = −Aω² cos ωt gives a = −ω²x, the SHM condition.

Maximum values

Quantity	Maximum	Where
Speed	v_{max} = Aω (when sin ωt = ±1)	at the **equilibrium position** (x = 0)
Acceleration	a_{max} = Aω² (when cos ωt = ±1)	at the **extremes** (x = ±A)

- The period of SHM **does not depend on the amplitude** (it is isochronous).

WORKED EXAMPLE

Amplitude and maximum acceleration

A loudspeaker cone vibrates in SHM at 240 Hz with a maximum speed of 2.25 m s^{-1}.

ω = 2πf = 2π × 240 = 1508 rad s^{-1}

A = v_{max}/ω = 2.25/1508 = 1.49 × 10^{-3} m = **1.49 mm**

$a_{max} = A\omega^2 = 1.49 \times 10^{-3} \times 1508^2 = \mathbf{3.39 \times 10^3\ m\ s^{-2}}$

WORKED EXAMPLE

Displacement and velocity at a given time

A mass oscillates with A = 4.0 cm and T = 0.80 s, released from x = +A at t = 0. Find x and v at t = 0.30 s.

$\omega = 2\pi/T = 2\pi/0.80 = 7.85$ rad s^{-1}; $\omega t = 7.85 \times 0.30 = 2.36$ rad (calculator in **radians**)

$x = A \cos \omega t = 4.0 \times \cos 2.36 =$ **−2.8 cm** (2.8 cm on the other side of equilibrium)

$v = -A\omega \sin \omega t = -0.040 \times 7.85 \times \sin 2.36 =$ **$-0.22\ m\ s^{-1}$** (moving in the −x direction)

EXAM TIP

Convert first: frequency "per minute" to Hz, cm or mm to m. Use **radians** for ωt. For maxima you don't need t at all: $v_{max} = A\omega$, $a_{max} = A\omega^2$.

MARK-SCHEME WORDING

- "Use of $\omega = 2\pi/T$" (or $2\pi f$) is its own mark; "use of $v = A\omega \sin \omega t$ with $\sin \omega t = 1$".
- Maximum KE: "use of $E_k = \frac{1}{2}mv^2$" with $v_{max} = A\omega$; or $E_{k,max} = \Delta E_{el} = \frac{1}{2}kA^2$ "because energy is conserved" (WPH15 Oct 2020).

Often asked as: amplitude or v_{max} from f; heart-beat rate per minute to Hz then amplitude; maximum KE from T and A; reading v_{max} from a graph; MCQs on $v_{max} = 2\pi A/T$ and phase.

COMMON MISTAKE

Calculator in degrees. Confusing f and T (or f and ω). Not seeing that maximum speed means $\sin \omega t = 1$. Giving an answer in m when mm was asked. Not squaring 2π when rearranging.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Speed at any displacement: **$v = \pm\omega\sqrt{(A^2 - x^2)}$**, which comes from energy conservation (150). Useful when you know x but not t.
- Acceleration is in **antiphase** with displacement (half a cycle out); velocity is a quarter cycle out of phase with both (WPH15 Jan 2021).

SPEC 145 Periods of a mass-spring system and a simple pendulum

WHAT THIS SPEC POINT ASKS

Use T = 2π√(m/k) for a mass on a spring and T = 2π√(l/g) for a simple pendulum.

Mass on a spring

Mass–spring oscillator

FORMULA SHEET

$$T = 2\pi\sqrt{\frac{m}{k}}$$

m: oscillating mass / kg k: spring constant / N m^{-1}

Period of a mass on a spring

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A mass m on a spring of stiffness k feels a restoring force $F = -kx$. Newton's second law, $F = ma$, gives

$$ma = -kx \quad \Rightarrow \quad a = -\frac{k}{m}x$$

2. This has the form $a = -\omega^2 x$, so the motion is SHM with

$$\omega^2 = \frac{k}{m} \quad \Rightarrow \quad \omega = \sqrt{\frac{k}{m}}$$

3. Use $T = 2\pi/\omega$:

$$\boxed{T = 2\pi\sqrt{\frac{m}{k}}}$$

A heavier mass has more inertia, so it oscillates more slowly. A stiffer spring pulls back harder, so it oscillates faster. The amplitude doesn't appear at all.

- m = oscillating mass (kg); k = spring constant (N m⁻¹). From ω² = k/m and T = 2π/ω.
- Heavier mass → longer period; stiffer spring → shorter period. T ∝ √m: four times the mass doubles T.
- The same T applies horizontally or vertically (gravity only shifts the equilibrium position).
- Find k from a static test: k = F/Δx = mg/Δx.

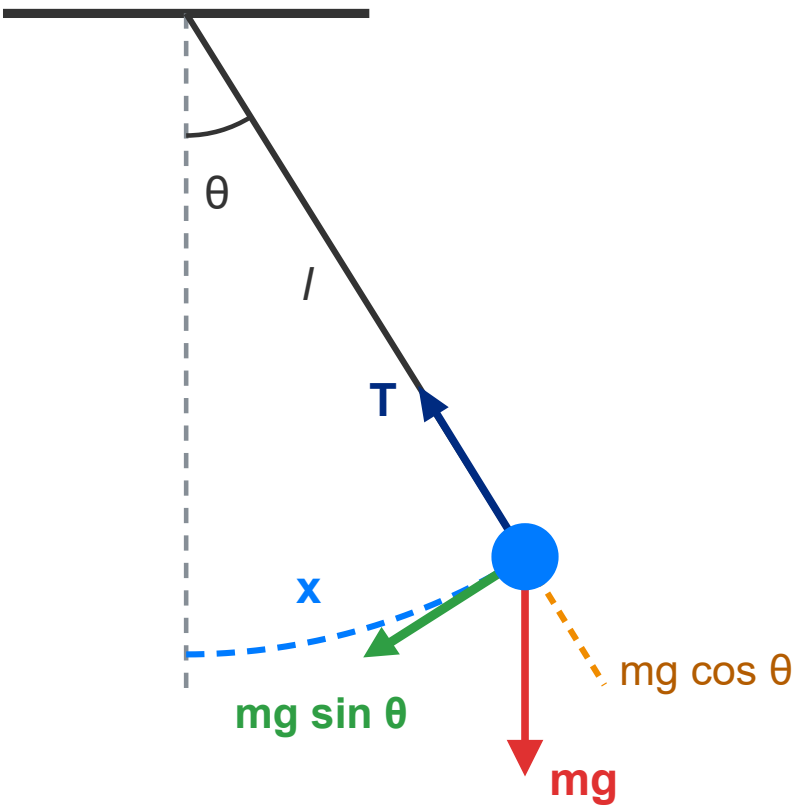
Simple pendulum (small angles)

Simple pendulum

FORMULA SHEET

$$T = 2\pi\sqrt{\frac{l}{g}}$$

l: length to the centre of mass of the bob / m g: 9.81 N kg⁻¹



The component of the weight along the arc, $mg\sin\theta$, pulls the bob back towards equilibrium. The tension and $mg\cos\theta$ act along the string and don't change its speed along the arc.

Period of a simple pendulum (small angles)

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A bob of mass m on a string of length l is pulled aside by an angle θ. The component of its weight along the arc pulls it back towards equilibrium:

$$F = -mg\sin\theta$$

2. For **small angles** (in radians), $\sin\theta \approx \theta$. The bob's displacement along the arc is $x = l\theta$, so $\theta = x/l$:

$$F \approx -mg\theta = -\frac{mg}{l}\,x$$

At 10°, $\sin\theta$ and θ differ by only 0.5%, so this works well for swings of up to about 10°.

3. Divide by m to get the acceleration. The mass cancels:

$$a = -\frac{g}{l}\,x \quad \Rightarrow \quad \omega^2 = \frac{g}{l}$$

4. Use $T = 2\pi/\omega$:

$$\boxed{T = 2\pi\sqrt{\frac{l}{g}}}$$

The period depends only on the length and on g: not on the mass of the bob, and not on the amplitude while the swing stays small.

- l = length from the pivot to the **centre of mass** of the bob (m); g = 9.81 N kg^{-1}.
- T depends only on **l and g**: **not** on the mass of the bob and not on the amplitude (as long as the angle is small, under about 10°).
- For a pendulum, ω² = g/l.

Getting k or g from a straight-line graph

Plot	Straight line because	Gradient gives
T² against m	T² = (4π²/k) m	k = 4π²/gradient
T² against l	T² = (4π²/g) l	g = 4π²/gradient

Plot	Straight line because	Gradient gives
f^2 against $1/m$	$f^2 = (k/4\pi^2)(1/m)$	$k = 4\pi^2 \times$ gradient

WORKED EXAMPLE

A seconds pendulum

Find the length of a pendulum with T = 2.000 s (g = 9.81 N kg⁻¹).

$l = gT^2/4\pi^2 = 9.81 \times 2.000^2/(4\pi^2) =$ **0.994 m**

It can't be used to define the metre exactly: **g varies with location**.

WORKED EXAMPLE

Extra mass on a spring

A 0.250 kg mass on a spring has T = 0.60 s. What mass must be added to make T = 0.75 s?

$k = 4\pi^2 m/T^2 = 4\pi^2 \times 0.250/0.60^2 = 27.4$ N m⁻¹

New total mass $m = kT^2/4\pi^2 = 27.4 \times 0.75^2/(4\pi^2) = 0.391$ kg

Added mass = 0.391 − 0.250 = **0.14 kg**

(Ratio shortcut: $m_2 = m_1(T_2/T_1)^2 = 0.250 \times (0.75/0.60)^2 = 0.391$ kg.)

WORKED EXAMPLE

A 0.40 kg mass hangs on a spring with $k = 60\ \text{N m}^{-1}$. It is pulled down 5.0 cm and released at $t = 0$. Calculate (a) the period, (b) the maximum speed, (c) the maximum acceleration, and (d) the displacement and velocity at $t = 0.10\ \text{s}$.

Show solution

(a) $T = 2\pi\sqrt{m/k} = 2\pi\sqrt{0.40/60} = 0.513\ \text{s}$, and $\omega = \sqrt{k/m} = \sqrt{150} = 12.2\ \text{rad s}^{-1}$.

(b) $v_{\max} = A\omega = 0.050 \times 12.25 = 0.61\ \text{m s}^{-1}$

(c) $a_{\max} = A\omega^2 = 0.050 \times 150 = 7.5\ \text{m s}^{-2}$

(d) $\omega t = 12.25 \times 0.10 = 1.225\ \text{rad}$ (radians!), so

$$x = A\cos\omega t = 0.050\cos(1.225) = 0.017\ \text{m} = 1.7\ \text{cm}$$

$$v = -A\omega \sin \omega t = -0.612 \times \sin(1.225) = -0.58 \text{ m s}^{-1}$$

The minus sign means it is moving back towards (and about to pass through) the equilibrium position, which makes sense: 0.10 s is just under a quarter of a period.

WORKED EXAMPLE

A student times 20 oscillations of a pendulum of length 0.800 m and gets 35.9 s. Calculate a value for g.

Show solution

$T = 35.9/20 = 1.795$ s. Squaring $T = 2\pi\sqrt{l/g}$ and rearranging:

$$g = \frac{4\pi^2 l}{T^2} = \frac{4\pi^2 \times 0.800}{1.795^2} = 9.80 \text{ N kg}^{-1}$$

A better method is to use several lengths and plot T^2 against l: the gradient is $4\pi^2/g$.

EXAM TIP

Substitute numbers first, then rearrange: most lost marks come from dropping the 2π or the square root while rearranging symbols. Time **many** oscillations (e.g. 20 or 30) and divide, rather than timing one.

MARK-SCHEME WORDING

- "Use of T = 2π√(m/k)" with the **total** oscillating mass; "allow use of ω² = k/m with T = 2π/ω".
- "Use of T = t_{mean}/30" for repeated timing.
- Two springs side by side sharing the load: "factor of 2 applied to either m or k" (WPH15 Jun 2023).
- Pendulum period doesn't depend on mass: adding mass to the bob changes nothing (except through l) (WPH15 Jan 2025).

Often asked as: pendulum length for a given T; pendulum on another planet (find g from g = GM/r² first); k from a load test then f; mass added to change T; k from an f² against 1/m graph.

COMMON MISTAKE

Stopping at the total mass when the question asks for the mass **added**. Forgetting the factor 2 for two springs. Using orbital equations when the question is about a **pendulum** on a planet (it scored zero). Rearranging before substituting.

⚠ Beyond the spec · not part of the Edexcel IAL specification

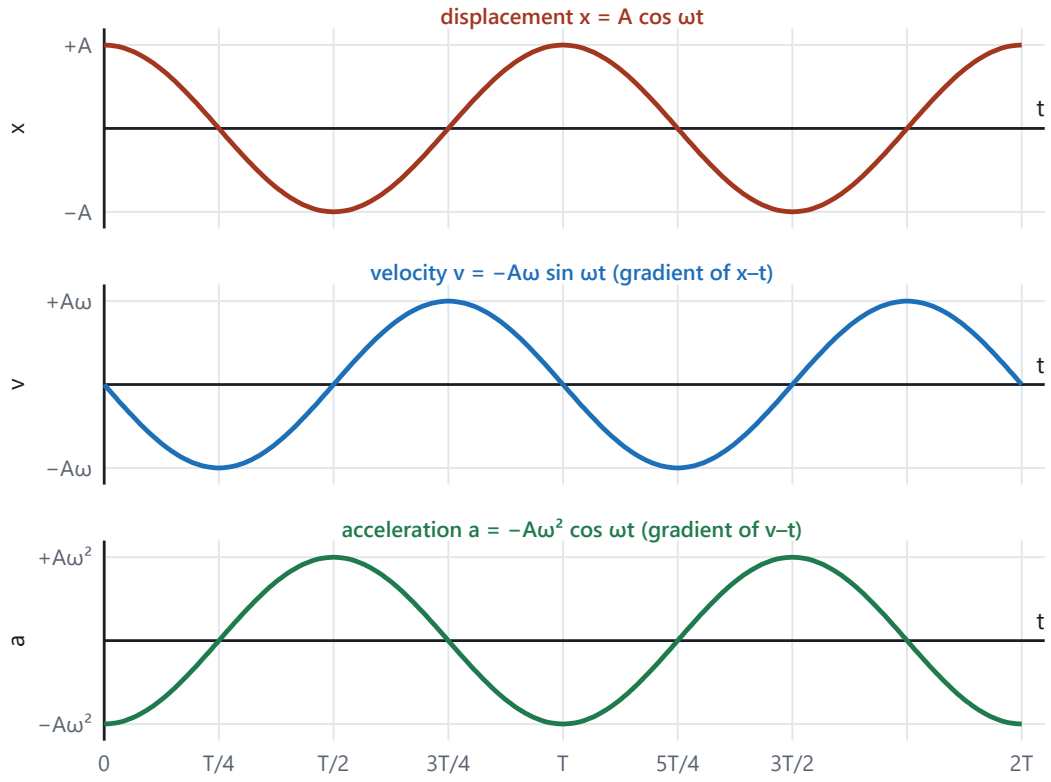
Examined in past papers even though the spec doesn't list it.

- **Springs in parallel** share the load, so the effective k doubles; in series the effective k halves (WPH15 Jun 2023).
- The maximum load a spring can take is smaller when oscillating than at rest, because at the lowest point the spring force exceeds the weight (WPH15 Oct 2021).

SPEC 146 Displacement-time graphs

WHAT THIS SPEC POINT ASKS

Draw and interpret a displacement-time graph for an oscillating object, knowing that its gradient at any point gives the velocity.



SHM starting from x = +A. Each graph is the gradient of the one above. v is a quarter cycle out of phase with x; a is exactly opposite to x (antiphase).

Reading the x–t graph

- **Amplitude** A = the height of a peak above the equilibrium line.

- **Period** T = time for one complete cycle (peak to next peak). Read it over several cycles and divide, for accuracy. Then f = 1/T and ω = 2π/T.
- **Gradient = velocity**:
  - at a **peak or trough** the gradient is **zero**: the object is momentarily at rest at the extreme;
  - where the graph **crosses the time axis** the gradient is **steepest**: maximum speed, at the equilibrium position.
- To find v at a point, draw a tangent and find its gradient (use a large triangle).

Sketching

- Use a sine or cosine shape depending on where timing starts: x = A cos ωt from an extreme, x = A sin ωt from the equilibrium position, x = −A sin ωt if it starts at equilibrium moving in the −x direction.
- Keep the **amplitude constant** (if undamped) and the period the same as any other graph given. Put **numbers on the time axis** (e.g. T/2, T or the actual times).

WORKED EXAMPLE

Maximum speed from a displacement-time graph

An x–t graph shows A = 3.0 cm and three complete cycles in 1.20 s.

T = 1.20/3 = 0.40 s

$v_{max} = A\omega = 2\pi A/T = 2\pi \times 0.030/0.40 =$ **0.47 m s^{-1}**

Check: the tangent where the graph crosses the axis should have a gradient of about 0.47 m s^{-1} (answers 0.45–0.55 m s^{-1} were accepted from a tangent) (WPH15 Jun 2025).

MARK-SCHEME WORDING

- Sketch from a v–t graph: "same time period as velocity and constant amplitude"; "wave shifted a quarter cycle" (WPH15 Jun 2021).
- "Minus sine graph", "approximately the same period", amplitude approximately constant (WPH15 Oct 2025).

Often asked as: sketch x–t to match a given v–t or a–t graph; read A and T; v_{max} from the gradient at the axis crossing.

COMMON MISTAKE

Leaving the time axis without values. Drawing the amplitude changing when there is no damping. Shifting by half a cycle instead of a quarter. Timing from one cycle only.

SPEC 147 Velocity-time graphs

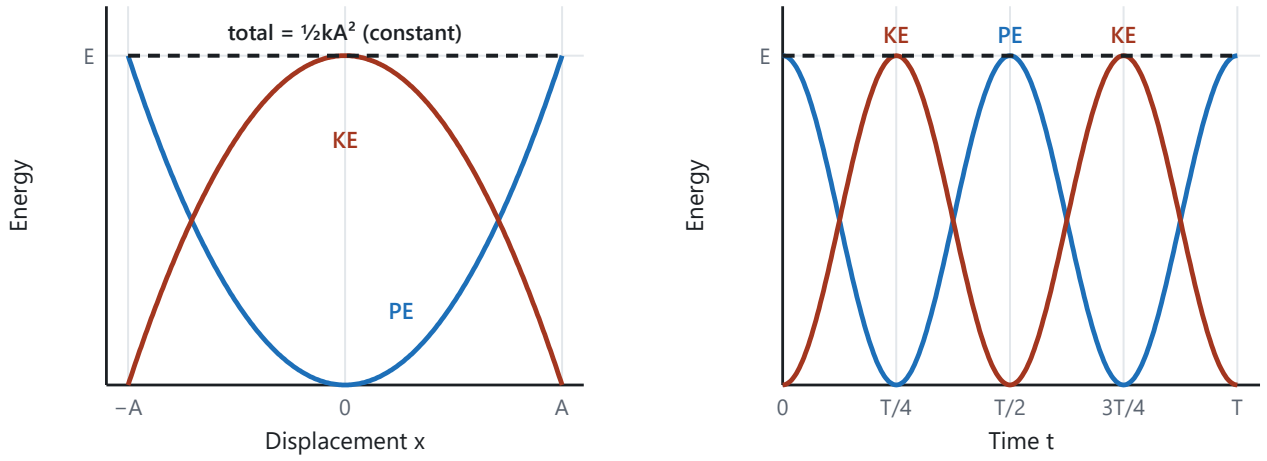
WHAT THIS SPEC POINT ASKS

Draw and interpret a velocity-time graph for an oscillating object, knowing that its gradient at any point gives the acceleration.

Reading the v–t graph

- The v–t graph is the gradient of the x–t graph: for $x = A\cos\omega t$ it is $v = -A\omega\sin\omega t$, a quarter cycle out of phase with x. Its peaks are at $\pm A\omega$.
- **Gradient = acceleration**:
  - where v crosses zero (object at an **extreme**) the v–t gradient is **steepest**: maximum acceleration, $\pm A\omega^2$;
  - at a **peak** of v (object at **equilibrium**) the gradient is **zero**: no acceleration, because there is no resultant force at x = 0.
- The **area** under a v–t graph from one extreme to the next gives the distance travelled, 2A.
- The a–t graph (gradient of v–t) is $a = -A\omega^2\cos\omega t$: the mirror image of the x–t graph.

Energy-time graphs



Energy in undamped SHM. Against displacement (left), PE is a ∪-shaped parabola, KE an ∩-shaped one, and they add to a constant total. Against time (right), each energy is always positive and goes through two cycles for every oscillation.

- $KE = \frac{1}{2}mv^2$, so KE is always **positive** and peaks every time the object passes through equilibrium, which happens **twice per cycle**. Energy–time graphs have **twice the frequency** of the displacement.

MARK-SCHEME WORDING

- Kinetic energy against time: "positive", "frequency double that of the displacement", "maximum at t = 0 when the mass passes the equilibrium position" (WPH15 Oct 2020).
- Energy against displacement is a **parabola** (WPH15 Jan 2025).

Often asked as: MCQ on the a–t graph from the gradient of a v–t graph; sketch KE against time; sketch energy against displacement.

COMMON MISTAKE

For energy against displacement, drawing a sine curve, a straight line, an upside-down parabola for PE, or negative energies. For KE against time, drawing it at the same frequency as x.

SPEC 148 Resonance

WHAT THIS SPEC POINT ASKS

Understand what resonance is.

The idea

- Every oscillating system has a **natural frequency** f_0: the frequency it oscillates at when displaced and left alone.
- When a periodic **driving force** acts at a frequency equal to the natural frequency, energy is transferred to the system most efficiently (each push comes at the same point in each cycle, in step with the motion). The amplitude builds up to a **maximum**.

Resonance: when the **driving (forcing) frequency equals the natural frequency** of the system, there is **maximum transfer of energy** to the system and the amplitude of oscillation is a **maximum**.

Examples

Useful	Harmful (must be damped or avoided)
A child on a swing pushed once per swing	A bridge driven by people walking or by wind (the London Millennium Bridge)
Rocking a car stuck in mud, in time with its natural rocking	Buildings in an earthquake whose natural frequency matches the ground's shaking

Useful	Harmful (must be damped or avoided)
Musical instruments: air columns and strings resonate	Car parts or panels rattling at a particular engine speed
Tuning a radio circuit to one station's frequency	A wine glass shattered by a sound at its natural frequency

- **Barton's pendulums**: a heavy driver pendulum hangs from a string with several light pendulums of different lengths. The one with the **same length** as the driver (same natural frequency) swings with the largest amplitude.

MARK-SCHEME WORDING

- "The **driving (forcing) frequency equals the natural frequency** of [the system named]", "**maximum transfer of energy**" (accept "efficient transfer"), "resulting in maximum amplitude".
- Name both frequencies for the system in the question: "driving frequency of the hand equals the natural frequency of the mass-spring system" (WPH15 Jun 2023: very few did).
- Wine glass (6-mark) (WPH15 Jun 2022): striking gives a **free** oscillation that dies away (damping); a wet finger sliding round the rim **drives / forces** the glass; resonance when the driving frequency equals (is close to) the natural frequency; large amplitude, so loud sound.
- Bridge (6-mark) (WPH15 Jan 2025): people force the bridge to oscillate; "large amplitude of oscillation" (not "large swaying"); energy transferred from the bridge to the dampers.

Often asked as: define resonance in a context (tide in a bay, a hand driving a spring, an animal on a spring balance, a carriage on a bumpy road); decide whether a situation is resonance (microwave at 2.45 GHz against a natural frequency near 10 GHz: no); bridge and building design.

COMMON MISTAKE

Calling every oscillation resonance: striking a glass is a free oscillation, not resonance. Saying "the frequencies match" without saying **which** frequencies, of **what**. Writing only "maximum amplitude" when the mark needs "maximum energy transfer". Assuming resonance is always bad.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Resonance can still occur when the driver pushes at a **sub-multiple** of the natural frequency (every second or third cycle), because each push still comes at the same point in the cycle (WPH15 Jan 2021: insect wing muscles).
- Not every "frequency" heating is resonance: a microwave oven runs at 2.45 GHz, well away from the natural frequency quoted for water molecules (near 10 GHz), so the answer to "is this resonance?" was **no** (WPH15 Oct 2020). Check the numbers before saying resonance.

- At resonance the energy supplied per cycle equals the energy lost to damping per cycle, so the amplitude stops growing (WPH15 Jun 2023).

SPEC 149 Core practical 16: unknown mass from resonant frequencies

WHAT THIS SPEC POINT ASKS

Find an unknown mass by measuring the resonant frequencies of a spring loaded with known masses.

Principle

- A mass m on a spring has natural (resonant) frequency f = 1/T = (1/2π)√(k/m), so

Resonant frequency and mass

$$f^2 = \frac{k}{4\pi^2} \cdot \frac{1}{m} \qquad \text{or} \qquad T^2 = \frac{4\pi^2}{k}\, m$$

- Measure f for several known masses, plot a calibration graph (f² against 1/m, or T² against m), then measure f for the unknown mass and read its mass off the graph.

CORE PRACTICAL

Core practical 16: method

1. Hang a spring from a clamp. Attach its bottom end to a known mass, and connect a **vibration generator** driven by a **signal generator** to the top of the spring (or below the mass).
2. Slowly increase the frequency from low values and watch the amplitude. Find the frequency at which the amplitude is **largest**: the **resonant frequency**. Approach it from above and below and average; use the signal generator's display or an oscilloscope to read f.
3. Repeat for 5–6 known masses (e.g. 100 g to 600 g).
4. Plot f² (y) against 1/m (x): a straight line through the origin with gradient k/4π².
5. Hang the unknown mass, find its resonant frequency, and read 1/m from the graph (or calculate m = gradient/f²).

Alternative: displace each mass and let it oscillate freely. The natural frequency equals the resonant frequency, so timing 20 oscillations (T = t/20, f = 1/T) gives the same data. Plot T² against m.

Variables: independent = mass; dependent = resonant frequency (or period); control = same spring, small amplitude, same clamp.

Safety: clamp the stand to the bench (or weight the base); wear eye protection in case the spring snaps; keep feet clear of falling masses.

WORKED EXAMPLE

Reading the unknown mass

A T²–m graph for a spring is a straight line through the origin with gradient 1.97 s² kg⁻¹. The unknown mass oscillates with T = 0.85 s.

k = 4π²/gradient = 39.5/1.97 = 20 N m⁻¹

m = T²/gradient = 0.85²/1.97 = **0.37 kg**

WORKED EXAMPLE

A student's graph of T^2 against m has a gradient of 1.58 s² kg⁻¹. An unknown mass on the same spring resonates at 1.40 Hz. Calculate (a) the spring constant and (b) the unknown mass.

Show solution

(a) The gradient is $4\pi^2/k$, so

$$k = \frac{4\pi^2}{1.58} = 25.0 \text{ N m}^{-1}$$

(b) $T = 1/f = 1/1.40 = 0.714$ s, so $T^2 = 0.510$ s^2 and

$$m = \frac{T^2}{\text{gradient}} = \frac{0.510}{1.58} = 0.32 \text{ kg}$$

Errors and improvements

- Resonance peaks are hard to judge by eye: find the frequency range where the amplitude is largest, take its midpoint, and repeat.
- The spring's own mass adds to the oscillating mass, so a T²–m line may not pass exactly through the origin. That doesn't stop the method working, because the unknown is read off the same line.
- Keep amplitudes small so the spring obeys Hooke's law.
- For timing: start and stop at the **equilibrium position** (fastest point, using a fiducial marker); time many oscillations.

MARK-SCHEME WORDING

- "Combine $T = 2\pi\sqrt{(m/k)}$ with $f = 1/T$ to obtain $f^2 = (k/4\pi^2)m^{-1}$", "compare with $y = mx + c$", "large triangle used", "conclusion consistent with the earlier value" (WPH15 Jun 2021).
- "Use of $T = t/30$" when 30 oscillations were timed (WPH15 Jun 2023).

Often asked as: the procedure itself hasn't been asked yet; the f^2 against $1/m$ analysis and repeated timing have. A data logger's advantage: many readings over very short or very long times (WPH15 Jun 2021).

COMMON MISTAKE

Reading T from a single oscillation. Plotting f against m and expecting a straight line. Forgetting that the gradient of f^2 against $1/m$ is $k/4\pi^2$, not k.

SPEC 150 Energy in undamped and damped oscillations

WHAT THIS SPEC POINT ASKS

Apply conservation of energy to undamped and damped oscillating systems.

Undamped: energy swaps between KE and PE

- With no energy losses, the total energy is **constant**. At the extremes it is all **potential** (elastic PE for a spring, gravitational PE for a pendulum); at the equilibrium position it is all **kinetic**.
- For a mass on a spring (taking PE = 0 at equilibrium):

Total energy of a mass on a spring

$$E_{\text{total}} = \tfrac{1}{2}kA^2 = \tfrac{1}{2}mv_{\max}^2$$

taking the potential energy as zero at equilibrium

Energy at displacement x

$$\tfrac{1}{2}mv^2 + \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2$$

Total energy of a mass–spring oscillator

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. The elastic potential energy stored in a spring is $\frac{1}{2}F\Delta x$. With $F = kx$, at displacement x it stores

$$E_\mathrm{p} = \tfrac{1}{2}kx^2$$

2. At the end of the swing, $x = A$ and the mass is at rest, so all the energy is potential:

$$\boxed{E_\mathrm{total} = \tfrac{1}{2}kA^2}$$

3. Check at equilibrium, where all of it is kinetic. Using $v_\mathrm{max} = A\omega$ and $\omega^2 = k/m$:

$$\tfrac{1}{2}mv_\mathrm{max}^2 = \tfrac{1}{2}mA^2\omega^2 = \tfrac{1}{2}mA^2\frac{k}{m} = \tfrac{1}{2}kA^2$$

The two agree, as energy conservation says they must. The energy is proportional to A^2: double the amplitude and the oscillator holds four times as much energy.

- **Energy ∝ amplitude²**. Double the amplitude and the energy goes up four times; halve the energy and the amplitude falls to 1/√2 (71%).

WORKED EXAMPLE

Speed from energy

A 0.20 kg mass on a spring (k = 50 N m⁻¹) oscillates with amplitude 4.0 cm. Find the total energy, v_{max}, and the speed at x = 2.0 cm.

E = ½kA² = ½ × 50 × 0.040² = **0.040 J**

v_{max} = √(2E/m) = √(2 × 0.040/0.20) = **0.63 m s⁻¹** (check: ω = √(k/m) = 15.8 rad s⁻¹, Aω = 0.63 m s⁻¹)

At x = 0.020 m: PE = ½ × 50 × 0.020² = 0.010 J, so KE = 0.040 − 0.010 = 0.030 J

v = √(2 × 0.030/0.20) = **0.55 m s⁻¹**

WORKED EXAMPLE

For the 0.40 kg mass on the 60 N m⁻¹ spring with amplitude 5.0 cm, find (a) the total energy and (b) the speed of the mass when its displacement is 3.0 cm.

Show solution

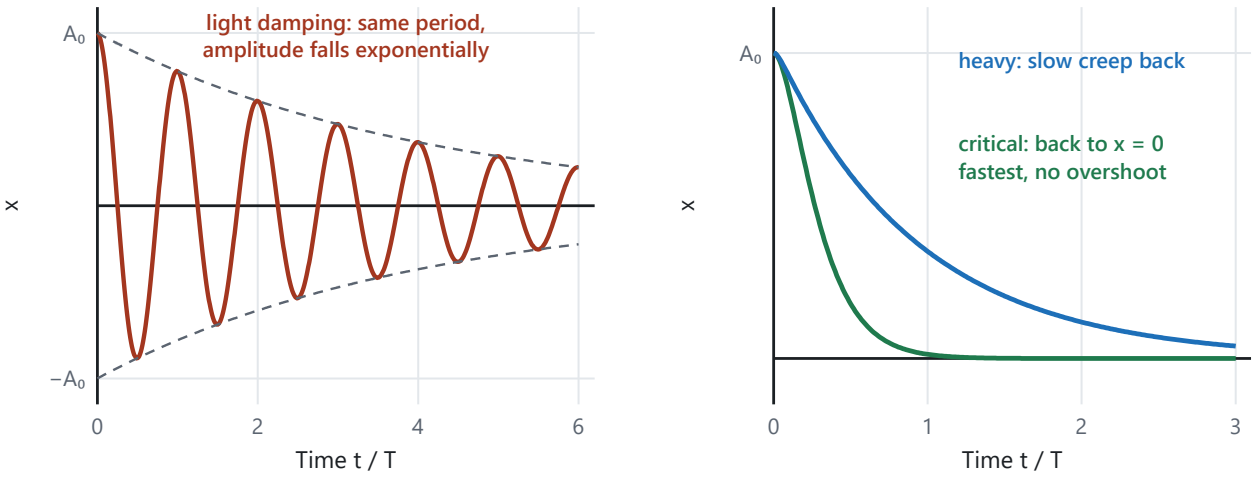
(a) $E = \frac{1}{2}kA^2 = 0.5 \times 60 \times 0.050^2 = 0.075$ J

(b) At $x = 0.030$ m: $E_\text{p} = 0.5 \times 60 \times 0.030^2 = 0.027$ J, so $E_\text{k} = 0.075 - 0.027 = 0.048$ J.

$$v = \sqrt{\frac{2E_\text{k}}{m}} = \sqrt{\frac{2 \times 0.048}{0.40}} = 0.49 \text{ m s}^{-1}$$

Damped: energy leaves the system

- **Damping**: **resistive forces** (air resistance, friction, a liquid's drag) act on the oscillator. **Work is done against** them, so energy is **transferred out of the oscillating system** to the **surroundings** (their thermal energy store rises).
- The total energy falls, so the **amplitude falls** with time. With light damping the period stays almost the same.
- The damping force always acts **opposite to the velocity** (it is not the restoring force, which acts opposite to the displacement).



Left: light damping. The period is unchanged; the amplitude falls by the same fraction each cycle (an exponential decay). Right: critical damping returns to equilibrium in the shortest time without overshooting; heavy damping returns slowly without oscillating.

Degree of damping	Behaviour	Example
Light	oscillates, amplitude decays gradually	a pendulum in air

Degree of damping	Behaviour	Example
Critical	returns to equilibrium in the shortest time, no oscillation	car suspension, a measuring needle
Heavy	returns very slowly, no oscillation	a door closer

MARK-SCHEME WORDING

- Damping: "energy is transferred (dissipated) to the surroundings" (or "the thermal energy store of the dampers increases") "because **work is done against resistive forces**" (WPH15 Jun 2024, Jun 2025, Jun 2026). Only "MAX 1 for reference to damping" without the work-done idea (WPH15 Oct 2023).
- "Work is done (by the roller) as oil is forced through the holes" (damper in a building) (WPH15 Jun 2024).
- Undamped: "elastic PE converted to KE at the equilibrium position, total energy constant".

Often asked as: maximum KE from elastic PE; draw a horizontal total-energy line on an energy–displacement graph and read off energies; why a mass oscillating in water stops; MCQs on amplitude $\propto \sqrt{E}$ and on the direction of the damping force.

COMMON MISTAKE

Explaining damping without "work done against resistive forces". Confusing the damping force with the resultant (restoring) force. Saying KE is constant, or zero at equilibrium.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Total energy = $\frac{1}{2}kA^2$, the area under the force–extension graph up to the amplitude (WPH15 Oct 2020, Jan 2023).
- The damping force is always opposite to the **velocity** (WPH15 Oct 2023, Oct 2025 MCQs).

SPEC 151 Free and forced oscillations

WHAT THIS SPEC POINT ASKS

Distinguish free oscillations from forced oscillations.

The difference

	Free oscillation	Forced oscillation
How it starts	displaced once (or struck), then left alone	a **periodic external (driving) force** keeps acting
Frequency	the system's **natural frequency** f_0	the **driving frequency** (once it has settled)
Energy	no energy supplied after the start; with damping, the amplitude dies away	energy is supplied by the driver each cycle, replacing the energy lost to damping
Examples	a plucked guitar string; a struck tuning fork; a pendulum pulled aside and released	a swing pushed every cycle; a car body vibrating as it drives over a bumpy road; the cone of a loudspeaker

- A forced oscillator first goes through a short messy stage, then settles to a steady amplitude at the driving frequency. How big that amplitude is depends on how close the driving frequency is to f_0 (spec point 152).

MARK-SCHEME WORDING

- Free: oscillates "at its natural frequency" after "a single displacement", with no external periodic force.
- Forced: "an external periodic force" makes it oscillate "at the driving frequency".
- Pushing a swing each time it reaches maximum height is a **forced** oscillation (WPH15 Oct 2025).

Often asked as: classify situations (MCQ); the striking/rubbing wine-glass 6-mark question (WPH15 Jun 2022).

COMMON MISTAKE

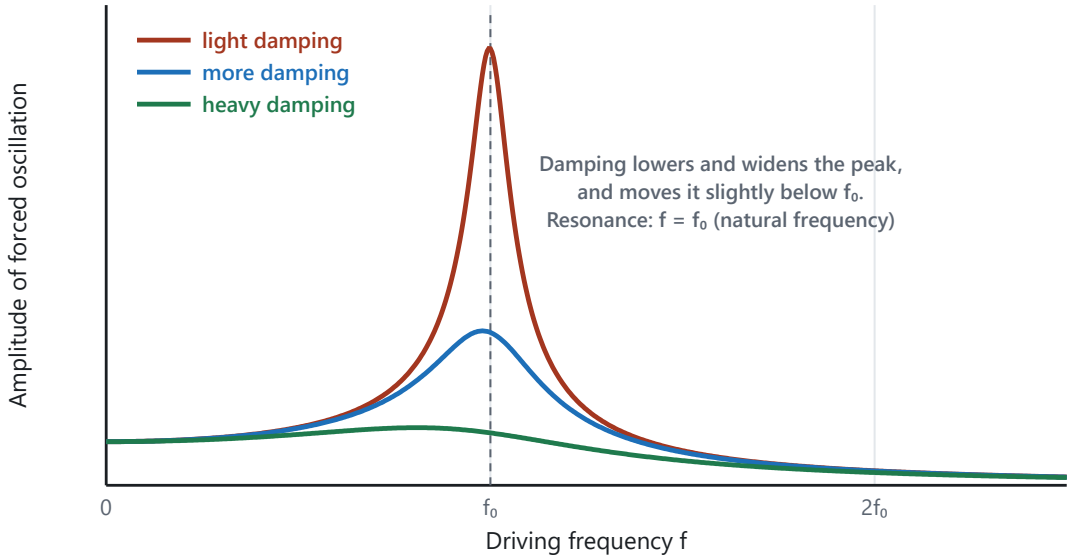
Describing both the struck glass and the rubbed glass as resonance: striking is free; rubbing is forced (and can produce resonance). Defining "forced" as "not free" without mentioning the external periodic force or the energy it supplies.

SPEC 152 Forced oscillations near the natural frequency; damping and resonance

WHAT THIS SPEC POINT ASKS

Understand how the amplitude of a forced oscillation varies with driving frequency, especially at and near the natural frequency, and know qualitatively how damping changes resonance.

The resonance curve



Amplitude of a forced oscillation against driving frequency. The peak is at (or just below) the natural frequency f_0. More damping gives a lower, broader peak, shifted slightly to lower frequency.

- **Low driving frequency**: the system just follows the driver; amplitude ≈ the driver's own amplitude (the static displacement).
- **As f approaches f_0**: amplitude rises sharply; maximum at resonance ($f \approx f_0$).
- **Above f_0**: amplitude falls, towards zero at very high frequency (the system can't keep up).

Effect of damping

- More damping means more energy is removed each cycle, at **every** frequency, so:
  - the **peak amplitude is lower**;
  - the peak is **broader** (flatter, less sharp), so the system responds over a wider range of frequencies;
  - the peak moves **slightly below** f_0.
- Heavy damping can remove the peak almost entirely.
- Damping **doesn't change the natural frequency** itself (very little, for light damping).

Why it matters

- Buildings in earthquake zones and long bridges are fitted with **dampers** (oil-filled pistons, tuned mass dampers, rubber bearings) so that if they are driven near their natural frequency the amplitude stays small. The damper is designed so that **energy is transferred efficiently into the damper**, where work done against resistive forces dissipates it.
- Where a sharp resonance is wanted (radio tuning, musical instruments), damping is kept small.

WORKED EXAMPLE

A tall building in an earthquake zone has a natural frequency of 0.8 Hz. Explain why it could be badly damaged by an earthquake with ground vibrations of about 0.8 Hz, and why engineers fit dampers.

Show solution

The ground shaking is a periodic driving force, so the building performs forced oscillations. When the driving frequency (0.8 Hz) equals the building's natural frequency, resonance occurs: energy is transferred to the building most efficiently, so the amplitude of its sway becomes very large and can exceed what the structure can withstand.

Dampers dissipate energy from the oscillations (for example by plastic deformation of ductile metal, or by friction in fluid dampers). This makes the resonance peak much lower, so the maximum amplitude is smaller, and it makes the oscillations die away quickly once the shaking stops.

MARK-SCHEME WORDING

- "Damping reduces the amplitude (of the peak)" and "the peak is flatter / broader"; energy is removed "at all frequencies" (WPH15 Jun 2021).
- Damper design: "maximum energy transfer to the damper" (from the building) (WPH15 Jun 2024).

Often asked as: MCQs on where damping acts; designing dampers for a building or carriage; describe how amplitude changes as the driving frequency is increased. A sketch of the curves hasn't been asked in recent papers, but learn it.

COMMON MISTAKE

Saying damping changes the natural frequency a lot, or moves the peak to a higher frequency. Drawing heavier-damping curves with taller peaks. Designing a building damper to "avoid resonance" when the question wants energy transferred into the damper.

SPEC 153 Damping by plastic deformation of ductile materials

WHAT THIS SPEC POINT ASKS

Understand how damping and the plastic deformation of ductile materials reduce the amplitude of oscillations.

Ductile materials

- A **ductile** material can undergo **large plastic deformation** (it can be drawn into wires) before it breaks. Examples: copper, mild steel, gold.
- **Plastic** deformation is **permanent**: the material doesn't return to its original shape when the force is removed.

How they damp oscillations

- When a ductile component (e.g. a steel brace or damper in a building) is deformed plastically, **work is done** to rearrange its structure permanently. The energy goes into its **internal energy** (it warms up) and is **not returned** to the oscillation as kinetic or elastic energy.
- So energy is **removed from the oscillating system** and the **amplitude falls**.
- An elastic material, by contrast, stores the energy as elastic PE and gives most of it back, so it doesn't damp much.
- Any damping (air resistance, viscous oil dampers, rubber bearings) works the same way: work done against resistive forces removes energy from the oscillation.

Earthquake engineering

- Buildings use **ductile** steel frames and connections, which bend permanently and absorb energy in a severe earthquake rather than snapping (as a **brittle** material such as unreinforced concrete or cast iron would).
- Other dampers: oil-filled hydraulic dampers, rubber base isolators, tuned mass dampers at the top of tall buildings.
- Related energy-absorbing ideas: car crumple zones, climbing ropes that stretch and absorb the energy of a fall.

MARK-SCHEME WORDING

- "Ductile materials deform **plastically**", so "energy is dissipated / work is done" in permanent deformation. A ductile material is one that shows "large plastic deformation before it breaks"; it is not "stiff" or "elastic" (WPH15 Jun 2023, Jan 2026).
- Steel is ductile (WPH15 Oct 2024).

Often asked as: MCQs only so far (which describes a ductile material; why ductile materials reduce amplitude), set in the context of buildings in earthquake zones.

COMMON MISTAKE

Confusing ductile with brittle, or with elastic. Saying the energy is "lost" without saying it is transferred out of the oscillating system (to internal energy of the material and the surroundings).

Quick checks

QUICK CHECK

Which equation describes an object moving with simple harmonic motion?

- $a = \omega^2 x$
- $a = -\omega^2 x$ ✓
- $a = -\omega x^2$
- $a = -g$

The acceleration must be proportional to the displacement (x, not x^2) and opposite to it, hence the minus sign. $a = -g$ is a constant acceleration, like a bouncing ball in the air, which is not SHM.

QUICK CHECK

An object in SHM passes through its equilibrium position. Which statement is correct?

- Its velocity is zero and its acceleration is a maximum.
- Both its velocity and its acceleration are a maximum.
- Its speed is a maximum and its acceleration is zero. ✓
- Both its velocity and its acceleration are zero.

At $x = 0$, $a = -\omega^2 x = 0$: there is no restoring force there. All of the energy is kinetic, so the speed is at its greatest, $A\omega$.

QUICK CHECK

A pendulum clock is taken to the Moon, where g is one sixth of its value on Earth. By what factor does the period change?

- It becomes 6 times longer.
- It becomes about 2.4 times longer.
- It becomes about 2.4 times shorter.
- It doesn't change, because the period doesn't depend on mass.

$T = 2\pi\sqrt{l/g}$, so $T \propto 1/\sqrt{g}$. Dividing g by 6 multiplies T by $\sqrt{6} = 2.45$. The period doesn't depend on the bob's mass, but it does depend on g.

QUICK CHECK

The amplitude of a mass–spring oscillator is doubled. What happens to its period and to its total energy?

- The period doubles and the energy doubles.
- The period is unchanged and the energy doubles.
- The period is unchanged and the energy is four times bigger.
- The period doubles and the energy is four times bigger.

$T = 2\pi\sqrt{m/k}$ has no A in it. The total energy is $\frac{1}{2}kA^2$, so doubling A multiplies it by 4.

QUICK CHECK

The damping of a driven oscillator is increased. How does its resonance curve (amplitude against driving frequency) change?

- The peak becomes taller and narrower.
- The peak becomes lower and broader.
- The peak moves to a much higher frequency.
- The curve doesn't change, only how quickly it is reached.

More damping dissipates more energy per cycle, so the amplitude at resonance is smaller and the response is spread over a wider range of frequencies. The peak also moves slightly *below* the natural frequency, not above it.

QUICK CHECK

A mass on a spring has a period of 0.50 s. The mass is replaced by one four times as heavy. What is the new period?

- 0.25 s
- 0.50 s
- 1.0 s
- 2.0 s

$T \propto \sqrt{m}$, so four times the mass gives $\sqrt{4} = 2$ times the period: 1.0 s.